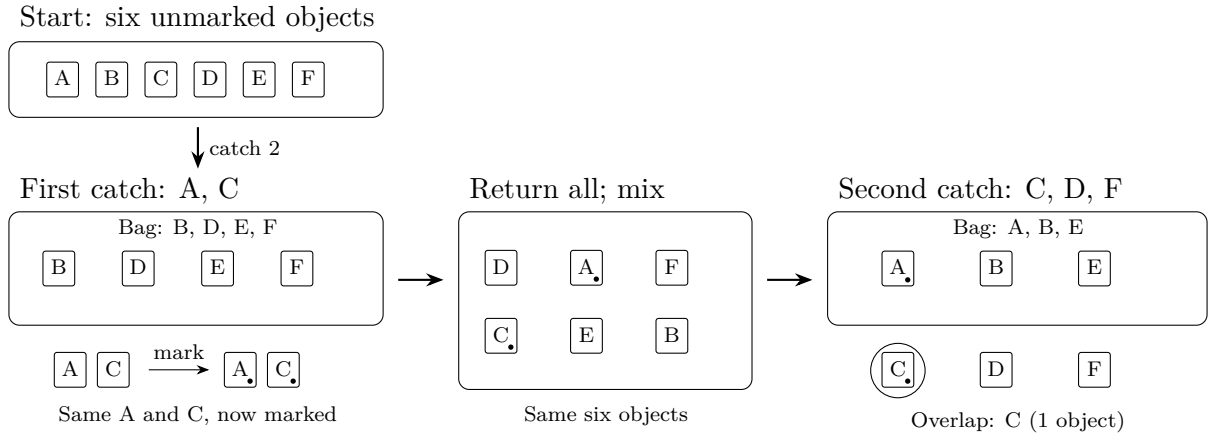
Use identical-size cards or counters with visible marker dots that cannot be felt. Keep the same objects throughout a series; add, remove or exchange none except for temporary catches. For sampling trials, draw without looking; every group of the chosen size should have the same chance, whether marked or unmarked.

Keep each catch out until it is counted. Mark the first catch, return *all* of it, and mix before the second catch. Marks stay on through both catches. An overlap is a marked object in the second catch. Before a new round, return everything and clear the marks. To show a record is possible, you may arrange catches: use distinct objects in each catch, from the same bag.



Problem 5: Make an 8-object bag and a 16-object bag. Arrange legal catches of 4 from each. Can either bag give both an overlap of 0 and an overlap of 3? Use your objects to settle it.

Problem 6: Use each bag below. The first catch has 5 objects and the second has 4. Find every overlap that can happen. Arrange catches to show your possibilities.

Bag total	Possible overlaps
6	
8	
12	

Problem 7: Choose a bag of 6, 7 or 8 objects. Make two catch records with first catch 5 and second catch 4: one that your bag could give and one that it could never give. Tell a partner the bag's total and the two records. Can they find the impossible record?